
Analyzing the Impact of Border Conditions on Adversarial Patch Attack Performance

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Abstract

Object detection models are vulnerable to adversarial patch attacks, and various studies have explored methods to distort model predictions using these attacks. However, prior research has primarily focused on optimizing the internal patterns of patches, while the role of patch borders has received little attention. In this study, we investigated how the presence of borders affects the recognition process of CNN-based object detection models using activation maps and GradCAM, and found that borders could influence the attack success rate (ASR). We evaluated adversarial patches under varying border conditions, including border thickness, color, and interpolation methods, to assess their impact on attack performance. Additionally, we found that a border thickness of approximately 3% (4-8 pixels) of the patch size is effective in increasing ASRs, with an improvement of up to 9.8%. Furthermore, we observed that using a gray border consistently enhanced the attack success rate in nearly all cases.

1 INTRODUCTION

Adversarial patch attacks are techniques designed to distort the performance or cause malfunctions in deep learning models, particularly visual recognition models (Brown et al., 2017; Eykholt et al., 2018). These attacks involve the application of patches to digital images or real-world objects (Liu et al., 2018b). These patches are designed to prevent the correct recognition of objects by exploiting the vulnerabilities of deep learning models, thereby achieving successful attacks.

(Song et al., 2018). Such patches can lead to unreliable results and pose significant risks in critical applications such as autonomous vehicles or security systems. These patches are typically generated as follows. First, the image of the target object to be attacked is analyzed to identify the optimal pattern that will cause the model to malfunction when the patch is applied (Williams and Li, 2024; Den Hollander et al., 2020). This pattern is adjusted at the pixel level to prevent the object from being accurately recognized. To ensure robustness across different environments, patches are optimized for multiple factors such as brightness, angle, and size (Kim et al., 2021).

However, the process of applying these patches to images is often inconsistent. For instance, when applying patches to digital images, black or white borders may appear around the patch (Tang et al., 2024; Sharma et al., 2022). In real-world environments, inconsistent borders are used, or thick borders are deliberately added to avoid covering the patch by hand (Thys et al., 2019; Wang et al., 2021).

The presence of borders in adversarial patch attacks can significantly influence the effectiveness of attacks on CNN-based models. Activation map results from the first convolution layer of the YOLOv5 model showed that the patch’s border was more strongly activated compared to other elements in the image, likely because the early layers of CNNs are typically composed of edge-detection filters. This activation propagates through deeper layers and may ultimately affect the performance of the patch attack. Additionally, Grad-CAM analysis revealed that when the patch had a border, the model tended to focus more on the patch, which could increase the ASR. These findings suggest that the presence and characteristics of borders can significantly influence the performance of adversarial patch attacks, indicating the need for further research under various conditions.

Our study aimed to analyze how border characteristics (presence, color type, and thickness) and interpolation methods affect attack success rates when patches are applied to background images. Moreover, we sought to

demonstrate that the application process and borders are critical factors in the effectiveness of the attack.

We conducted experiments on patches that perform hiding attacks, which prevent object detection models from detecting an object, and altering attacks, which cause one object to be recognized as another. During the experiments, we examined how the thickness and color of the patch borders affected the performance of the attacks. For instance, we analyzed how the results varied with thicker borders or specific colors. Furthermore, we applied different interpolation methods that can be used when applying patches and investigated how each interpolation method impacts the effectiveness of the patch. These experiments provided a comprehensive evaluation of how patch borders and interpolation methods impact ASRs. Finally, we also examined how the changes according to the application ratio affected the outcomes.

In summary, the primary contributions of this work can be outlined as follows:

- We highlight the lack of consideration for patch borders in existing adversarial patch research and emphasize the necessity to maintain border consistency.
- We investigate the impact of border presence on the recognition process of CNN-based object detection models subjected to adversarial patch attacks.
- We analyze the impact of patch border thickness, color, and various interpolation methods on the ASR and explore patch border conditions that can improve the ASR.

2 BACKGROUND

2.1 Adversarial Patch Attack

Adversarial patches are small image patterns that are designed to deceive machine learning models, particularly image classifiers or object detectors. These patches are created using various loss functions and optimization techniques to disrupt the predictions of the model.

The initial image for the patch is typically obtained by applying random noise or a neutral color (e.g., gray). To update and optimize the patch, various loss functions are defined. The main loss functions include: classification loss, which minimizes the difference from the target class, objectness score loss (Thys et al., 2019), which reduces the probability of detecting objects in object detection models, total variation (TV) loss (Mahendran and Vedaldi, 2015), which maintains

the visual naturalness of the patch, non-printability score (NPS) loss (Sharif et al., 2016), which ensures that the patch appears natural when printed or displayed, and saliency loss (Den Hollander et al., 2020), which encourages the patch to be placed in less important areas of the image.

The patch is then applied to the image with various transformations such as different positions, sizes, angles, and brightness values to enhance its robustness. The patched image is then fed into the model to obtain prediction results, and the values of the defined loss functions are calculated. The patch is updated to minimize the loss using gradient descent. This process is repeated until the patch is optimized. The final patch is evaluated on various images and scenarios to assess how effectively it deceives the target model. Through these steps, the generated adversarial patch is optimized to effectively deceive the target model in diverse environments.

Thus, loss functions and transformations applied during patch creation has primarily considered in studies on adversarial patch attacks. However, this paper focuses on differences in interpolation methods and border conditions that can occur when applying the patch.

2.2 Patch Attack for Vision Task

2.2.1 Classification

Brown et al. (2017) introduced adversarial patches, which deceive images without training data. These patches are effective but must be large and noticeable, with their effectiveness varying under different conditions. Karmon et al. (2018) proposed LaVAN, achieving high success rates by covering less than 2% of pixels without obscuring main objects; however, it is location-dependent. Liu et al. (2018a) suggested a method for generating more natural adversarial patches, but its performance fluctuates based on context. Zhou et al. (2018) introduced DiAP, which performs attacks without training data by manipulating intermediate model activations.

2.2.2 Object Detection

DPatch, proposed by Liu et al. (2018b), can attack both bounding box regression and object classification in object detection tasks. Lee and Kolter (2019) introduced a patch robust to real-world lighting and location changes. Madry et al. (2017) ensured attack effectiveness by maintaining pixel values within the RGB range. Song et al. (2018) proposed the first physical-world patch for object detection, enabling attacks to hide objects or create fake detections. Thys et al. (2019) developed effective patches against various hu-

man shapes, while Hollander et al. (2019) introduced camouflage patches to conceal military assets.

3 MOTIVATION

Research on patches has primarily focused on reducing their size or making them less noticeable. While prior studies have aimed to increase ASRs by adjusting the characteristics of the patches, they have not sufficiently considered the pixels and borders between the background image and patch.

Thys et al. (2019) proposed an attack that can significantly reduce the accuracy of human detection models by generating adversarial patches that target different shapes of people, and showed that it works effectively even when captured on camera in real-world situations. However, patches that are applied in both digital and real-world environments have noticeable white borders. Wang et al. (2021) demonstrated that, unlike a fixed digital patch, a patch in a physical environment works effectively even when it is rotated or conditions change. However, in this study, the physical patch was displayed on the screen of an electronic device (liquid-crystal display), with a wide black border around the patch. Similarly, the presence of black borders can be observed in the adversarial patch experiments conducted by Tang et al. (2024) and Sharma et al. (2022).

To investigate the impact of patch borders on attacks, we analyzed the YOLOv5 model’s response to images with borders from two perspectives. First, we extracted the activation map from YOLOv5’s initial convolution layer (Figure 1) and observed that the bordered patches exhibited significantly stronger activations than any other part of the image. This can be attributed to the fact that the initial layers of CNNs consist primarily of edge-detection filters, which respond strongly to sharp changes in pixel values. Given the strong activations observed in the early layers, we hypothesized that this information would become more abstract in deeper layers, potentially having a significant impact on the attack’s effectiveness.

Second, using Grad-CAM on YOLOv5’s final convolution layer (Figure 2), we found that the model focused more on the patch than on the person when the patch was bordered. Since the goal of the adversarial patch attack was “hiding,” this subtle shift in focus toward the patch could potentially improve the success rate of the attack. These results suggest that the performance of adversarial patch attacks on CNN-based models is highly influenced by the presence of patch borders. Further research considering various patch and border conditions is necessary. More detailed Activation Map and Grad-CAM results can be found in the Appendix. Moreover, given that CNN-based mod-

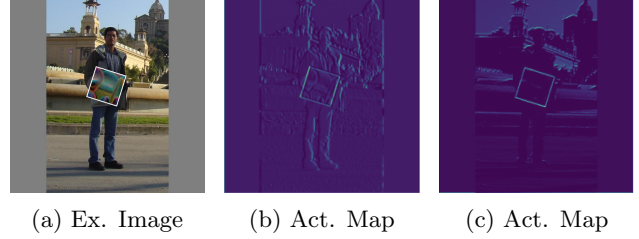


Figure 1: Activation Map of the First Layer.

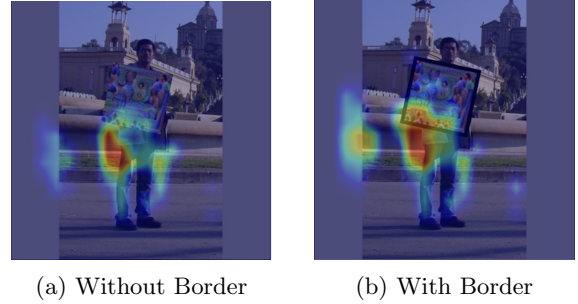


Figure 2: Grad-CAM of the Last Convolutional Layer.

els continue to be widely used in both research and real-world applications, this study is crucial for developing more effective methods for both attacking and defending such models.

4 EXPERIMENT

In this paper, we investigate the impact of borders on the ASR by conducting an experiment to attack human objects in a digital environment by applying an adversarial patch on them. Thus, we can assume that a human is holding an adversarial patch.

4.1 Experimental Settings

4.1.1 Datasets

In this study, we used the Microsoft Common Objects in Context (COCO) dataset (Lin et al., 2014) and the INRIA Person dataset (Dalal and Triggs, 2005) to train and evaluate our patches. The COCO dataset comprises 80 object categories covering a variety of everyday scenes and contains approximately 118,000 training and 5,000 validation images. We selected images that include the “person” class from the COCO validation set for our experiments. The INRIA dataset comprises human full-body data, consisting of 614 training and 288 test images. Using these datasets, we created four types of adversarial patches as shown in Table 1, which presents combinations of datasets used for patch training and attack validation.

In addition, We evaluated adversarial patch attacks us-

Table 1: Four Types of Patches

PATCH NAME	TRAIN DATASETS	ATTACK DATASETS
COCO_COCO	COCO	COCO
INRIA_COCO	COCO	INRIA
INRIA_INRIA	INRIA	INRIA
COCO_INRIA	INRIA	COCO

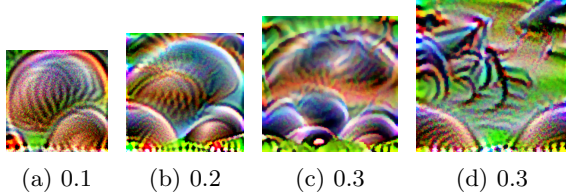


Figure 3: Examples of generated INRIA_INRIA hiding patches by attachment ratio.

ing two You Only Look Once (YOLOv5) models: one trained on the COCO dataset and the other trained on the INRIA dataset.

4.1.2 Patch Generation

The goal of the hiding attack was to prevent the YOLOv5 model from detecting humans. To achieve this, we generated and tested four types of patches: COCO_COCO, INRIA_COCO, INRIA_INRIA, and COCO_INRIA. The initial patch color was set to gray, and the loss functions comprised saliency loss, TV loss, NPS loss, and objectness score, as mentioned in Section 2.1. The loss weights were set to 1.0, 2.5, 0.1, and 3.0, respectively, to reduce the objectness score of the person class. Patch transformations were applied as referenced in Section 2.1. Examples of the INRIA_INRIA hiding patches that were generated with different application ratios are shown in Figure 3.

Altering attacks are classified into two types: targeted and untargeted. A targeted attack misleads the model to predict a specific target label, whereas an untargeted attack predicts one of several possible labels. We created untargeted patches that were predicted as various classes such as teddy bear, kite, and traffic light. The loss weights were set to 0.25, 0.5, 0.1, 8.0, and 6.0 for saliency loss, TV loss, NPS loss, original classification score, and target classification score, respectively, to reduce the classification confidence score for the person class and increase it for the target class. The target class was set to the one with the lowest classification score at each epoch. Other training conditions were the same as those for the hiding patches. Figure 4 shows examples of untargeted altering attack patches that were predicted as various classes.


Figure 4: Examples of generated altering patches by target class. From left to right, patch sizes are 256×256 , 128×128 , and 64×64 pixels, respectively.

4.1.3 Patch Sizes and Ratios

When generating adversarial patches, the patch size must be specified in pixel units. As the patch size increases, a greater variety of colors and shapes can be represented within the patch. Thus, the characteristics of the final patch can vary greatly depending on the initial size set during patch creation. Moreover, the attacker determine what proportion of the object the patch will cover during the application process. Generally, the larger the proportion of the patch on the object, the higher the ASR. Therefore, we investigate how the ASR changes in each scenario with and without borders by setting up experimental environments with various patch sizes and ratios.

4.2 Border Conditions

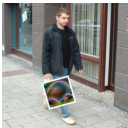
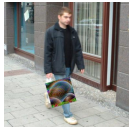
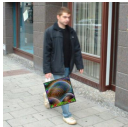
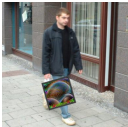
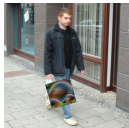
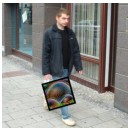
4.2.1 Interpolation

An important factor to consider when applying a patch is interpolation. Interpolation adjusts the image by calculating new pixel values to match the patch size to the object size in the original image (Amanatiadis and Andreadis, 2009; Han, 2013; Patel and Mistree, 2013; Parsania et al., 2014). Various interpolation methods have been developed such as linear, cubic, and nearest neighbor, each producing different border colors for the patch. This is because each method processes the pixel values at the patch edges differently.

Note that a binary mask is used to ensure proper alignment of adversarial patches with objects in images. This mask marks the area of the object with white pixels (value 1) and the external area with black pixels (value 0). During application, the patch is placed only on the white pixel areas, ensuring precise placement and minimizing any mismatch in pixel size or position.

The explanation of various interpolation methods and the resulting patch borders when applying adversarial patches using these methods are as follows. First, when the bilinear method is used, the unknown pixel value is calculated by using the values and distances

Table 2: Examples of patch applications according to color and thickness of the border.

COLOR	THICKNESS		
	2 pixels (0.78%)	8 pixels (3.13%)	16 pixels (6.25%)
White			
Gray			
Black			

of its four neighboring pixels. Since the binary mask of the adversarial patch sets the external area of the patch to zero, several zero-value pixels appear around the border, making it appear black. Second, when the bicubic method is used, the new pixel value is estimated by calculating a weighted average of its 16 nearest neighbors, with closer pixels having higher weights. Similar to the bilinear interpolation method, this also results in black borders due to the zero-value pixels in the binary mask. Third, when the nearest neighbor method is used, the value of the closest pixel is used directly without considering other pixels. This method fills the border with the closest pixel value, making the border less noticeable compared to bilinear or bicubic interpolation.

4.2.2 Color of the Borders

To analyze the effect of border conditions on ASRs, the border colors were varied as white, gray, and black. Examples of hiding patches with different border colors in a patch size of 256×256 pixels are shown in Table 2, where the first, second, and third rows show the patches with white, gray, and black borders, respectively.

4.2.3 Thickness of the Borders

The border thickness was maintained below 6.25% of the patch dimensions to ensure that the borders did not excessively influence the visual perception of the patch itself. For patches of 256×256 pixels, the border thicknesses were set to 2 (0.78%), 8 (3.13%), and 16 (6.25%) pixels. The results of applying patches with

different border thicknesses can be seen in each column of Table 2. Similarly, for patches of 128×128 pixels, the border thicknesses were set to 2 (1.56%), 4 (3.13%), and 8 (6.25%) pixels. For patches of 64×64 pixels, the border thicknesses were set to 2 (3.13%) and 4 (6.25%) pixels.

Various conditions that might occur when applying patches were considered and transformations were applied accordingly. Gaussian random noise was added, patches were rotated at varying angles, and they were applied to different locations on the object. Moreover, to simulate various scenarios, adjustments were made to the contrast and brightness values.

4.3 Evaluation Metric

The hiding patch attack was considered to be successful when objects labeled as “person” were misclassified as “background.” The hiding ASR is the proportion of detected “person” objects that were misclassified as “background” out of the total detected “person” objects in the dataset. The following equation was used to determine the hiding ASR:

$$ASR_{hiding} = \frac{N_{person \rightarrow background}}{N_{person}}, \quad (1)$$

where $N_{person \rightarrow background}$ represents the number of “person” objects that are misclassified as “background” and N_{person} denotes the total number of detected “person” objects.

The altering patch attack was considered to be successful when objects labeled as “person” were misclassified as a targeted altering class. The altering ASR is the proportion of detected “person” objects that were misclassified as the target altering class out of the total detected “person” objects in the dataset. To compute the altering ASR, we used the following formula:

$$ASR_{altering} = \frac{N_{person \rightarrow target}}{N_{person}}, \quad (2)$$

where $N_{person \rightarrow target}$ represents the number of “person” objects that are misclassified as the target altering class and N_{person} denotes the total number of detected “person” objects.

5 RESULTS

5.1 Impact of Interpolation Methods

When applying the adversarial patch to the object, the bicubic method applied the patch with a black border

Table 3: Comparison of ASRs by patch types and interpolation methods. The used patches are 256×256 in size hiding patches.

Ratio	0.1		0.2		0.3		0.1-0.3	
Interpolation	B	N	B	N	B	N	B	N
COCO_COCO	16.8	16.8	39.7	38.2	63.9	62.9	60.4	58.8
COCO_INRIA	45.2	44.8	66.9	64.7	92.8	91.6	92.3	91.3
INRIA_COCO	7.4	7.3	32.5	31.2	72.9	71.2	55.0	54.1
INRIA_INRIA	22.1	22.1	77.7	75.3	96.2	95.8	94.7	94.2

around it, whereas the nearest neighbor method applied the patch without a black border. In the binary mask of the adversarial patch, the area outside the patch was set to 0, resulting in several pixels around the border having a value of 0, resulting in a border with a black appearance. In the case of bilinear interpolation, the results were identical to those of bicubic interpolation. Table 3 lists the results of the different interpolation methods for different patches on the training and attack datasets. The patches used here had sizes of 256×256 pixels. We used the aforementioned four types of hiding patches and varied the percentage of patches that were applied to objects.

The results indicate that, regardless of the patch type, the ASR is higher when using bicubic interpolation. The difference was up to 2.4%, with the ASR ranging from 75.3% to 77.7% when INRIA_INRIA patch was applied with a ratio of 0.2, suggesting that the black border enhances the effectiveness of the patch in the detection model. This implies that even small changes in pixel values due to interpolation can significantly impact the ASR. Therefore, to maximize the effectiveness of an adversarial patch, adding a thin black border around the patch can be beneficial.

5.2 Impact of Patch Sizes and Ratios

Table 4 and Table 5 list the results of the hiding and altering patch attacks under various border conditions, respectively. We compared cases with and without borders; for cases with borders, we examined the differences based on patch size and patch ratio. In these tables, bold texts indicate the maximum increase in ASR compared to the no-border condition, while underlining indicate the minimum decrease in ASR compared to the no-border condition, or similar.

The ASR difference for hiding patches did not depend significantly on patch size. We defined “ASR difference” as the absolute value of the maximum difference between the case with no border and the cases with various borders. For example, when applying the INRIA_INRIA patch at a rate of 0.1, the ASR was 22.1% with no border on a patch of 256×256 pixels and 15.3% for a patch with a 16-pixel white border, exhibiting a

difference of 6.8%. Similarly, for a 128×128 patch, the ASRs were 21.7% and 17.2% for the patches with no border and 8-pixel white border, that is, a difference of 4.5%. For a 64×64 patch, the success rate was 22.6% without a border and 15.3% with a 4-pixel white border, a difference of 7.3%. Furthermore, the 0.2 and 0.3 ratios also show no trends with respect to patch size, suggesting that the differences in the ASRs with and without borders is not affected by patch size. The ASR difference of the altering patch also did not vary significantly with patch size.

The ASR difference between the hiding patch with and without borders varies by up to 9.8%, with an average of 4.5%, depending on the patch application ratio. Considering that ASR difference is positive in most cases, except for a few special instances, the variation in ASR difference is primarily interpreted as an increase in ASR. For the six patches tested with an application ratio of 0.1, the INRIA_INRIA 64×64 patch showed the highest absolute value of the maximum difference at 7.3%, with an average value across all patches of 4.5%. Similarly, with an application ratio of 0.2, the maximum and average values are 9.8% and 5.1%, respectively, while at a ratio of 0.3, they are 7.2% and 3.4%. In addition, for altering patches, ASR difference varies by an average of 2.4% depending on the patch application ratio. Generally, when applying adversarial patches to objects, they are applied at a size between 0.1 and 0.3 of the object size, suggesting that patch borders can help improve the ASR.

5.3 Impact of Border Thickness

We observed no consistent trend regarding the impact of border thickness on ASRs. In hiding attacks, INRIA_COCO patches show higher ASRs with thicker borders, whereas INRIA_INRIA patches exhibit lower ASRs with thicker borders. In altering attacks, INRIA_COCO (Kite) patches show no trend with respect to border thickness, whereas COCO_COCO (Traffic light) patches exhibit higher ASRs with thicker borders. Therefore, a consistent relationship between border thickness and ASR could not be identified.

However, in general, using the thickness of approximately 3% (4 and 8 pixels) clearly enhanced the ASRs, with an improvement of up to 9.8%. This specific thickness range consistently provided higher success rates under various conditions, making it a crucial strategic element to consider for improving ASRs.

5.4 Impact of Border Colors

For the INRIA_COCO hiding patch, the ASR generally increased when the border color was white compared to that in the unbordered state, with an max-

Table 4: Hiding ASRs under various patch conditions.

PATCH TYPE	PATCH SIZE	THICK-NESS	PATCH AREA RATIO								
			0.1			0.2			0.3		
			White	Gray	Black	White	Gray	Black	White	Gray	Black
INRIA-COCO	256 x 256	16	10.6	8.0	8.7	35.6	33.9	35.5	71.6	74.5	70.6
		8	10.0	7.9	8.3	35.5	34.0	<u>32.6</u>	70.8	72.8	<u>68.1</u>
		2	7.6	7.3	<u>7.0</u>	33.3	33.0	33.9	71.2	71.3	70.8
		None	7.3			31.2			71.2		
	128 x 128	8	8.5	<u>5.0</u>	5.9	32.1	29.4	30.3	66.9	66.4	<u>61.6</u>
		4	7.3	5.8	5.9	29.8	29.9	<u>28.0</u>	64.4	65.8	62.4
		2	6.9	6.9	7.3	30.2	30.5	28.7	63.9	63.7	62.1
		None	5.4			28.9			61.7		
	64 x 64	4	9.6	7.3	8.7	27.8	23.6	26.0	53.7	54.1	53.9
		2	8.3	<u>5.7</u>	6.8	24.6	<u>20.2</u>	23.3	50.0	<u>49.6</u>	51.0
		None	5.2			18.0			46.9		
INRIA-INRIA	256 x 256	16	<u>15.3</u>	21.9	18.5	76.2	80.2	76.8	<u>95.5</u>	96.2	96.2
		8	17.4	22.1	19.8	77.5	80.2	77.9	95.8	96.2	96.4
		2	20.0	21.5	21.7	78.9	79.2	78.1	96.2	96.2	96.4
		None	22.1			75.3			95.8		
	128 x 128	8	<u>17.2</u>	24.0	20.6	<u>68.5</u>	74.0	69.1	94.3	96.8	<u>94.2</u>
		4	18.9	22.5	22.3	68.7	74.3	71.5	94.9	97.0	96.0
		2	20.0	22.5	22.6	70.6	72.8	72.5	95.3	96.8	96.2
		None	21.7			71.7			96.4		
	64 x 64	4	<u>15.3</u>	22.1	20.6	60.0	64.2	<u>57.9</u>	<u>91.5</u>	92.8	92.1
		2	17.4	22.3	21.9	61.5	63.8	60.6	92.1	93.0	91.9
		None	22.6			63.0			91.1		

imum increase of 9.8% and an average increase of 4.5%. Conversely, when border color was black, ASR was similar or reduced. For the INRIA-COCO hiding patch, ASR generally increased when the border color was gray, with a maximum increase of 4.9% and an average increase of 1.5%. In contrast, when border color was white, ASR was similar or tended to decrease. Moreover, we found that selecting gray consistently resulted in an increase, regardless of the dataset.

These findings indicate that patch characteristics are influenced by the training dataset, reflecting the inherent properties of each patch type. The INRIA-COCO hiding patch was trained on the COCO dataset, while the INRIA-INRIA hiding patch was trained on the INRIA dataset. As each patch was designed to be inconspicuous within its respective training dataset, we can infer that the color characteristics of the dataset affect the change in ASR based on the border color. This trend can also be observed in Table 5. All the altering patches were trained with the COCO dataset and, similar to the hiding patches, showed an increase in ASR with white borders, whereas the ASR remained the same or was reduced for the patches with black borders. To maximize patch attack effectiveness, it is crucial to use border colors that align with the color characteristics of the training dataset.

Figure 5 shows example images of the hiding attack results according to border color differences. The images are displayed in order of confidence, from highest to lowest (i.e., from lowest to highest ASR). The patch used here is INRIA-INRIA, and the ASR is highest when the border is gray and tends to be lowest when the border is white.

The first row shows an example where a patch of size 128×128 at a ratio of 0.1 is applied to a person object with a border thickness of 8 pixels at a ratio of 0.1. The confidence score fall below 0.4 only when the border is gray, thus successfully performing a hiding attack; conversely, the confidence score when the border is white is 0.83, indicating the lowest attack performance among the failed attacks. The second row shows an example where a patch of size 128×128 at a ratio of 0.2 is applied to a person object with a border thickness of 8 pixels at a ratio of 0.2. In this case, the hiding attack is also successful only when the border is gray. Finally, the third row shows an example where a patch of size 128×128 at a ratio of 0.3 is applied to a person object with a border thickness of 2 pixels at a ratio of 0.3. In this case, the attack is successful in all cases except for the patch with the white border, and even in this case, the confidence score is significantly low at 0.45.

Table 5: Altering ASRs under various patch conditions.

PATCH TYPE	TARGET CLASS	PATCH SIZE	THICK-NESS	PATCH AREA RATIO					
				0.2			0.3		
				White	Gray	Black	White	Gray	Black
INRIA-COCO	Kite	512 x 512	32	4.3	4.8	<u>3.9</u>	11.5	12.2	<u>10.2</u>
			16	5.3	6.0	4.7	11.9	12.1	11.6
			8	4.3	5.7	4.4	12.2	12.1	11.8
			None	6.3			13.4		
COCO-COCO	Traffic Light	128 x 128	8	9.3	7.9	8.1	20.5	20.4	20.0
			4	8.3	7.8	7.3	21.1	20.5	20.2
			2	7.8	7.2	<u>7.2</u>	20.6	20.0	<u>19.7</u>
			None	6.8			19.4		



Figure 5: Examples of hiding attack results based on border color and ratio. In each row, patches are applied with ratios of 0.2 and 0.3 from the top to bottom rows. The numbers at the bottom of the images represent the object confidence scores, and the lower these values, the higher the ASR. The threshold was 0.4; therefore, scores below 0.4 are considered as a “Success” and scores above 0.4 are considered as a “Failed” attack.

6 CONCLUSION

This study highlights the often-overlooked importance of patch borders in adversarial patch research and underscores the necessity of maintaining consistent border conditions. Specifically, we input patch images with borders into YOLOv5 and visually analyzed them through Activation Maps and Grad-CAM, confirming that borders can significantly impact the recognition process of CNN-based object detection models. Furthermore, we conducted experiments to specifically analyze which patch conditions affect attack performance. The results showed that using bicubic interpolation increases the ASR. Additionally, we found that

using borders with a thickness of approximately 3% of the patch size can lead to an increase in ASR of up to about 9.8%, with gray borders enhancing the ASR on average. This suggests that it is crucial to adequately consider border conditions during patch design.

This study presents a new approach to enhancing the success rate of adversarial attacks, and future research would benefit from validating these strategies across various datasets, environmental conditions, and more complex models. Ultimately, this research will contribute to a deeper understanding of adversarial patch attacks and seek to develop advanced strategies for defending against them.

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